

## Part XII. COMPARE THE DOCUMENTED AND ARTIFICIAL MUTATIONS

|                                                | <b>WT Sequence</b>                                         | <b>Documented mutation</b>                                                                                            | <b>Student-created mutation</b>                                                                                                               |
|------------------------------------------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CDS length</b>                              | 1,086 bp                                                   | 1,248 bp                                                                                                              | 1,085 bp                                                                                                                                      |
| <b>Protein length</b>                          | 361 aa                                                     | 415 aa                                                                                                                | Altered/shortened                                                                                                                             |
| <b>Mutation type</b>                           | No mutation                                                | CAG trinucleotide repeat expansion                                                                                    | 1-nucleotide deletion (c.426delC)                                                                                                             |
| <b>Whether the reading frame changed</b>       | No                                                         | No                                                                                                                    | Yes                                                                                                                                           |
| <b>Whether a premature stop codon appeared</b> | No                                                         | No                                                                                                                    | Yes                                                                                                                                           |
| <b>Amino acids affected</b>                    | None                                                       | 54 additional glutamine (Q) residues                                                                                  | Many downstream amino acids are changed                                                                                                       |
| <b>Expected functional consequence</b>         | Normal ATXN3 protein sequence and expected normal function | Expanded polyglutamine tract may alter ATXN3 protein folding, aggregation, and cellular behavior associated with SCA3 | Likely severe disruption of the protein because the reading frame changes, producing altered downstream amino acids and premature termination |

The WT ATXN3 protein remained unchanged, while the documented mutation added 54 glutamine residues without shifting the reading frame. In contrast, the artificial 1-nucleotide deletion caused a frameshift, changing downstream amino acids and producing a premature stop codon.